

Dust Generator Budiman

For Dust Aerosol Generation during Inhalation Studies



General

The **TSE Dust Generator acc. to Budiman** has a robust design enabling prolonged, uniform generation of dust aerosol. The motivation to develop such a highly efficient particle generator is for their use in inhalation exposure experiments combined with either a Head Nose Only or Whole Body Exposure Unit.

Additional, flexibility has been integrated with the compatibility of two reservoirs enabling short or long exposures with high or low dust concentrations to be achieved.

All component parts of the instrument in contact with the dust, including the stability platform and scraper are fabricated from stainless steel, enabling this robust design, which in addition is easy to clean.

Application

The **TSE Budiman Dust Feeder** is suitable for most dry dusts that can be compressed to dry dust cakes. For compression the TSE pneumatic press can be used (optional).

The working principle of the Budiman Dust Feeder uses two variable parameters, controlled via a control unit:

- the advancement of the dust cake, and
- the rotational speed of the scraper.

The reservoir (small/ large containing the dust cake) is advanced on a stability platform towards the scraper, while the scraper is rotated on central axis. The dust is removed from the scraper down through the axis and dispersed by the high velocity flow inside the outlet nozzle, into the appropriate inhalation unit below.

Ordering Information

Cat. No.	Description
700700-SG-08	Dust Generator acc. to Budiman

Specifications subject to change without notice

Copyright © 2014 TSE Systems International Group - All rights reserved

Operation Modes

The dust generator can be used in either an automatic or manual mode. For automatic operation the dust generator can be connected via its own control unit to the system control unit. The dust concentration can be kept constant via the motor regulation or controlled via a loop function in conjunction with the **TSE Concentration Tendency- Measuring Unit** (700800-KM-01).

In manual operation the rotary knobs on the front panel of the dust generator control unit can be used to preset the speed at which the reservoir advances and the scraper rotates either separately or in a linked mode and thus influence the amount of dust aerosol generated.



Scraping blade (large)